

Assignment 2.2: Power Supply Systems Theory

Task 1: List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

SMPS:

Table 1. SMPS Use

Device	Use of SMPS
Desktop Computer	Converts AC power into stable DC voltages required by motherboard, CPU and other components
Laptop Charger	Converts AC power from the wall into low-voltage DC for charging the laptop battery
LED TV	Supplies efficient and regulated DC power to the TV's internal electronic circuits

UPS:

Table 2. UPS Use

Device	Use of UPS
Desktop Computer	Prevents sudden shutdown during power outages, protecting files and hardware too
Wi-Fi Router	Keeps the internet connection active during power cuts

Task 2: Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections
Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.

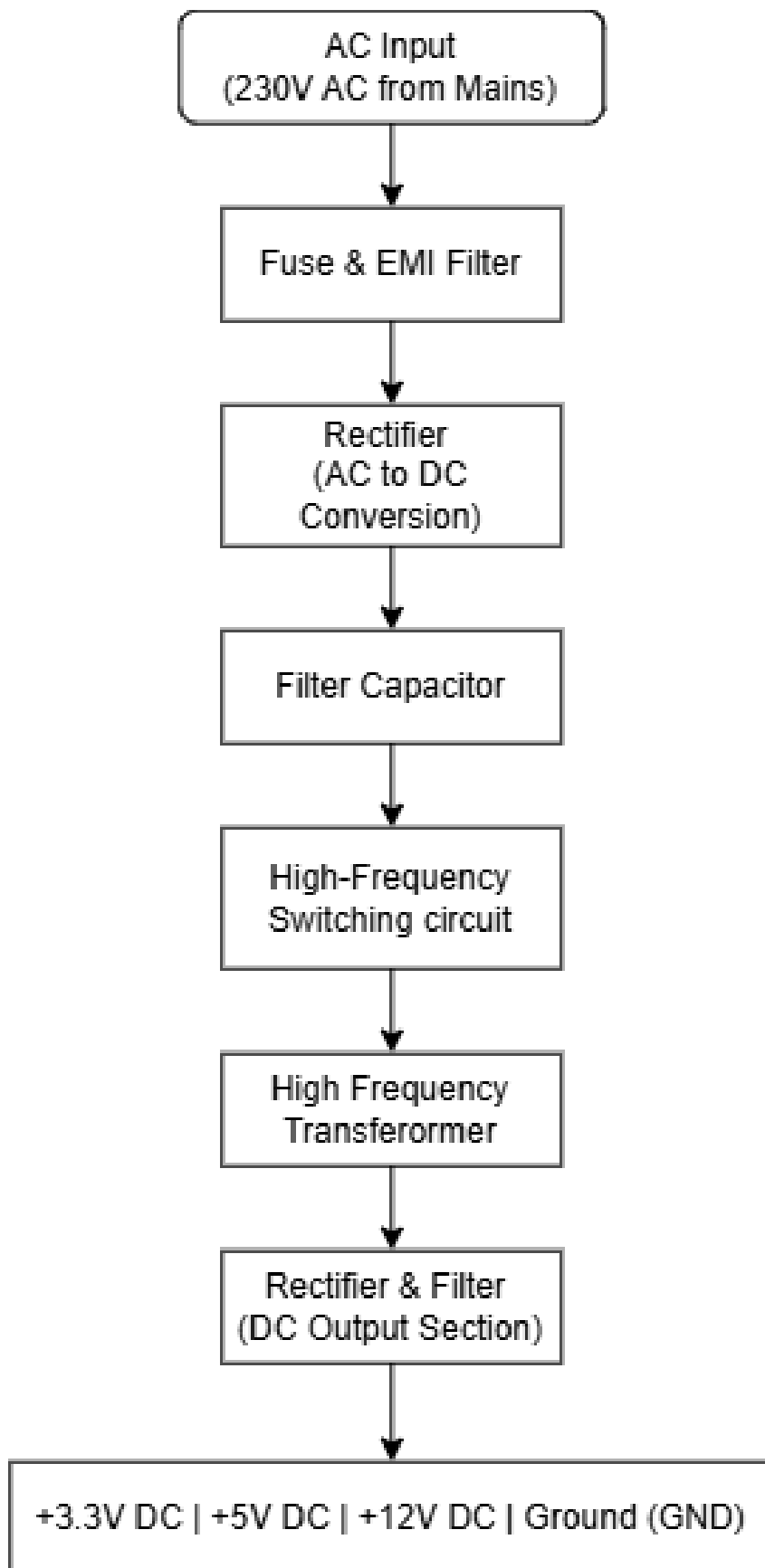


Fig 1. SMPS Components

Task 3: Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

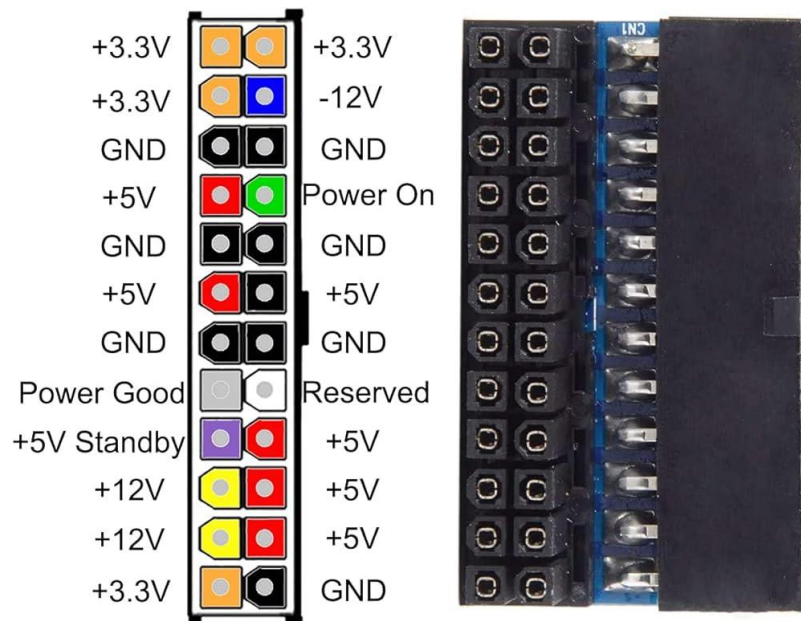


Fig 2: ATX-24-Pin motherboard connector

Device It Powers: Motherboard, Supplier power to the CPU, RAM, PCIe slots, and other onboard components.

Safety Tip: Turn off the power supply and unplug the power cable before connecting or disconnecting it. Ensure the connector clicks into place.

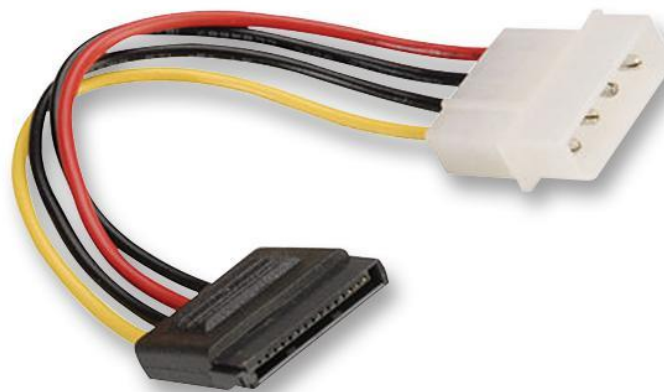


Fig 3: SATA Power connector

Device it Powers: Powers SATA devices such as SSDs, HDDs, and DVD devices.

Safety Tip: Align the L-shaped connector correctly and never force into the port.

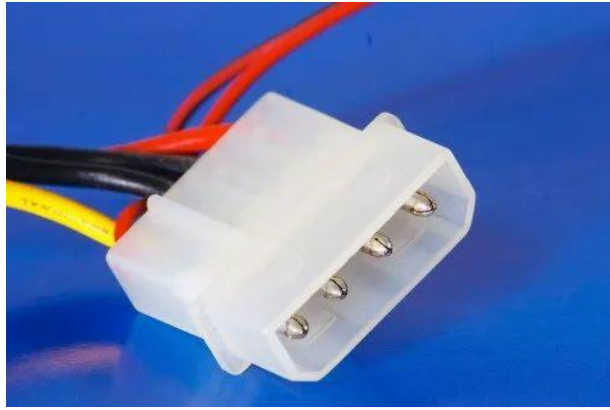


Fig 4. Molex Power Connector

Device it Powers: Powers older hard drives, optical drives, cooling fans, and some accessories.

Safety tip: Hold the plastic connector while unplugging; do not pull on the wires.



Fig 5. 6-pins to 8-pins PCI Connector

Device it Powers: Powers dedicated graphics card that require additional power.

Safety tip: Ensure the connector is fully inserted and locked before turning on the computer.

Task 4: Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Step 1: Install the SMPS

- Place the **SMPS (Power Supply Unit)** inside the computer case.
- Secure it with the provided screws.

Step 2: Connect the Motherboard

- Use the 24-pin ATX Power Connector.
- Plug it into the **24-pin ATX socket on the motherboard** until it clicks into place.

Step 3: Connect CPU Power

- Use the 8-pin (4+4) EPS/CPU Power Connector.

- Connect it to the CPU power socket near the processor on the motherboard.

Step 4: Connect the Graphics Card (GPU)

- Install the graphics card into the PCIe x16 slot on the motherboard.
- **Use the PCIe 6-pin or 8-pin Power Connector from the SMPS.**
- Connect it to the graphics card until the locking clip clicks.

Step 5: Connect Storage Devices

- Install the SSD or HDD in the drive bay.
- Connect the **SATA Power Connector** from the SMPS to the SSD/HDD.
- Connect a SATA Data Cable from the storage device to the motherboard.

Step 6: Connect Additional Devices (Optional)

- If needed, connect **Molex or SATA power** connectors to case fans, RGB controllers, or other accessories.

Step 7: Check All Connections

- Ensure all power connectors are firmly connected.
- Verify that cables are properly routed and do not block the cooling fans.

Step 8: Power On and Test

- Connect the main power cable to the SMPS.
- Turn on the SMPS switch (I position).
- Press the PC power button and check that the motherboard, graphics card, storage devices, and fans are working correctly.